

Proton Beam Therapy Leads to Excellent Local Control Rates in Choroidal Melanoma in the Intermediate Fundus Zone

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- **PURPOSE:** To evaluate long-term outcomes of proton beam radiotherapy in the treatment of choroidal melanoma of the intermediate zone of the fundus.
- **DESIGN:** Retrospective interventional single-center study.
- **METHODS:** The study was a retrospective analysis with long-term follow-up of 62 patients with a minimum tumor-to-disc and tumor-to-fovea distance of 2 mm of choroidal melanoma in the intermediate zone of the fundus. Mean values of tumor prominence, largest basal diameter, and tumor distances to the optic disc and fovea were 7.6, 12.8, 5.2, and 4.6 mm, respectively. All patients were irradiated with a total proton dose of 60 cobalt gray equivalents.
- **RESULTS:** After proton beam radiotherapy, 71.0% of the patients received subsequent endoresection of the tumor. Only 18 patients (29.0%) did not require additional tumor resection and were analyzed as a separate group. For the total of patients, the median follow-up time was 70.3 months. The 5-year Kaplan-Meier rates of local tumor relapse, enucleation, and distant metastasis were 3.9%, 3.7%, and 13.4%, respectively. Cataract surgery was the most frequent secondary treatment in our cohort.
- **CONCLUSIONS:** In this study we demonstrate the effectiveness of proton beam irradiation in tumor control and preservation of the globe in the analyzed patients. The rate of metastasis was not higher than previously described. Nevertheless, consecutive tumor surgery is frequently required to maintain the eye in patients with large-sized choroidal melanomas. In conclusion, proton beam radiotherapy allows preservation of the eye in mid-zone choroidal melanomas. (Am J Ophthalmol 2014;158:1184–1191. © 2014 by Elsevier Inc. All rights reserved.)

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ENUCLEATION FOR CHOROICAL MELANOMA HAS BEEN replaced by radiotherapy as an accepted treatment standard. The most popular techniques used for radiotherapy of choroidal melanoma are brachytherapy and proton beam therapy. There are currently more than 30 years of experience with proton beam irradiation.¹ The advantage of using protons for irradiation is the fact that the proton energy can be modulated to cover target tissue of any shape and at any depth with a nearly uniform dose owing to its physical characteristics, including minimal scatter and delivery of maximum dose at the end of the beam path, known as Bragg peak. Several investigators have demonstrated a noninferiority of proton beam irradiation with respect to metastasis and tumor-specific survival.^{2,3} While proton beam therapy has been advocated for central choroidal melanoma with close proximity to the disc and fovea, studies demonstrated that tumors exceeding 6 mm in height can be successfully treated with proton beam therapy if the inflammatory reaction after irradiation is met with surgical excision of the tumor tissue via a transscleral approach or endoresection.²

The present retrospective analysis presents the long-term outcomes in patients with proton beam irradiation for choroidal melanoma in the intermediate zone of the fundus at the Center for Ocular Oncology in Berlin.

METHODS AND MATERIALS

• **PATIENTS AND DATA COLLECTION:** This is a retrospective single-center study of patients with choroidal malignant melanoma treated with proton beam radiotherapy in Berlin. The study was approved by the institutional review board of the Charité–Universitätsmedizin Berlin, Germany, and was in accordance with the Declaration of Helsinki specifications. This study focuses on tumors in the intermediate zone of the fundus not attaching the ciliary body. The minimal distance of the unifocal lesion to both the optic disc and fovea was 2 mm. Consequently, exclusion criteria for this study were tumor invasion of the anterior eye segment, parapapillary location or location close to the fovea, and pretreatment of the tumor.

From 1998 until 2005, a total number of 62 patients met the inclusion criteria and received proton therapy. Forty-four